

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering **ASSESSMENT TOOL 1 – Data Interpretation**

Course Code:	ITA05	Course Title:	Computer Vision
CO Assessed:	CO4 – Articulation (combining methods for evaluation) P4	Assessment:	Assessment Tool 1 – Data Interpretation
Weightage:	40% of CIA	Total Marks:	100
CO4 Statement:	<i>Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL5)</i>		
Student Name:	S. Susannal Devapriyam	Reg. No.:	192525398
Section / Batch:	B. Tech AI & ML (2025)	Date:	17-08-2026

Code of Conduct

I, S. Susannal Devapriyam (192525398) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: S. Susannal Devapriyam.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data	
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided	
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data	
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions	
Clarity	10%	Presents interpretation clearly using proper structure, visuals.	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand	

Questions:

11. Detection Model Comparison under Resource Constraints

Performance:

Model Accuracy Latency

A 85% 40 ms

B 92% 90 ms

C 90% 60 ms

Compare the models and evaluate which is optimal for real-time systems with moderate accuracy requirements. Justify your decision.

12. Robustness vs Speed Trade-off

Performance:

Method Accuracy Robustness Speed

A 88% High Slow

B 85% Moderate Fast

C 90% Low Moderate

Compare the methods and evaluate the best choice for a balanced system. Justify trade-offs.

13. Optical Flow Stability Analysis

Results:

Method Accuracy Noise Sensitivity

A High High

B Moderate Low

Compare both methods and evaluate suitability for real-time tracking in noisy environments.

14. Motion Estimation Accuracy vs Efficiency

Results:

Method Error Computation

A 4% High

B 9% Low

C 6% Moderate

Compare methods and evaluate the best approach for embedded systems.

15. GMM Performance in Different Conditions

Results:

Scenario Accuracy

Static 92%

Dynamic 68%

Compare performance and evaluate whether GMM is reliable for all environments.

Suggest improvements.

11. Detection Model Comparison under Resource Constraints

Given Data

Model	Accuracy	Latency
A	85%	40 ms
B	92%	90 ms
C	90%	60 ms

1. Understanding of Data

The three detection models show a clear trade-off between **accuracy and latency**.

- **Model A** has the lowest accuracy of 85%, but it has the lowest latency of 40 ms.
- **Model B** provides the highest accuracy of 92%, but its latency is also the highest at 90 ms.
- **Model C** provides 90% accuracy with a moderate latency of 60 ms.

Thus, increasing accuracy generally comes with increased processing time.

2. Analysis

For real-time systems, latency is very important because the system must process input and produce a response quickly. A model with very high accuracy but excessive latency may not be suitable for applications such as:

- Real-time object detection
- Surveillance
- Autonomous systems
- Video analytics
- Industrial monitoring

Model A is the fastest, with only 40 ms latency. However, its 85% accuracy may be insufficient when the system requires reasonably accurate detection.

Model B has the highest accuracy at 92%, but its 90 ms latency makes it slower. It may be suitable for applications where accuracy is more important than response time.

Model C provides 90% accuracy while maintaining a relatively low latency of 60 ms. It therefore provides a better balance between detection quality and processing speed.

3. Application of Concepts

For a real-time system with moderate accuracy requirements, the objective is not necessarily to select the model with the highest accuracy. Instead, the model should provide sufficient accuracy while keeping latency low.

A useful comparison is:

- Model A → fastest but least accurate
- Model B → most accurate but slowest
- Model C → high accuracy with moderate latency

Model C achieves 5 percentage points higher accuracy than A while increasing latency by only 20 ms. Compared with B, Model C sacrifices only 2 percentage points of accuracy but reduces latency by 30 ms.

4. Conclusion

Model C is the optimal choice for a real-time system with moderate accuracy requirements.

It provides 90% accuracy and 60 ms latency, giving a good balance between performance and responsiveness. Although Model A is faster, its lower accuracy may reduce detection reliability. Model B is more accurate, but its 90 ms latency can affect real-time responsiveness.

Therefore, Model C offers the best accuracy–latency trade-off and is the most suitable overall choice.

12. Robustness vs Speed Trade-off

Given Data

Method	Accuracy	Robustness	Speed
A	88%	High	Slow
B	85%	Moderate	Fast
C	90%	Low	Moderate

1. Understanding of Data

The three methods demonstrate a trade-off between **accuracy, robustness, and speed**.

- **Method A:** 88% accuracy, high robustness, but slow.
- **Method B:** 85% accuracy, moderate robustness, and fast processing.
- **Method C:** Highest accuracy of 90%, moderate speed, but low robustness.

No single method performs best in all three characteristics.

2. Analysis

Method A has high robustness, meaning it can continue to perform reliably under changing or difficult conditions. Its 88% accuracy is also good. However, its main disadvantage is its slow processing speed.

Method B is the fastest method. This makes it useful for real-time applications. However, its accuracy is the lowest at 85%, and its robustness is only moderate.

Method C has the highest accuracy at 90% and moderate speed. However, its low robustness is a significant limitation. A method that performs well under ideal conditions may produce unreliable results when environmental conditions change.

Therefore, choosing only based on accuracy would incorrectly favour Method C.

3. Application of Concepts

A balanced system requires an appropriate compromise among accuracy, robustness, and speed.

Method	Main Strength	Main Weakness
A	High robustness	Slow
B	High speed	Lower accuracy
C	Highest accuracy	Low robustness

For a system where conditions may vary, robustness is important because the model should not lose performance when noise, illumination, motion, or input conditions change.

Method A provides 88% accuracy, which is only 2 percentage points lower than C, while offering high robustness. Therefore, the small loss in accuracy is compensated by improved reliability.

4. Conclusion

Method A is the best choice for a balanced system.

Although Method C has the highest accuracy, its low robustness can cause unreliable performance in practical environments. Method B is fast but sacrifices accuracy.

Method A offers good accuracy (88%) and high robustness, making it a more reliable overall solution. Its major disadvantage is speed, so optimization techniques such as model compression, hardware acceleration, or efficient implementation could be used to reduce its processing time.

Hence, Method A provides the most balanced trade-off between accuracy, robustness, and speed.

13. Optical Flow Stability Analysis

Given Data

Method	Accuracy	Noise Sensitivity
A	High	High
B	Moderate	Low

1. Understanding of Data

The two optical-flow methods have different performance characteristics.

- **Method A** provides high accuracy but is highly sensitive to noise.
- **Method B** provides moderate accuracy but has low sensitivity to noise.

Therefore, there is a direct trade-off between accuracy and robustness to noise.

2. Analysis

Optical flow is used to estimate the apparent movement of objects between consecutive frames.

In real-world video, noise can be introduced by:

- Camera vibration
- Poor lighting
- Sensor noise
- Motion blur
- Background changes
- Image compression

Method A's high accuracy is beneficial when the input video is clean. However, its high noise sensitivity means that its performance can degrade significantly when noise is present.

Method B has moderate accuracy but is less affected by noise. This makes it more stable under difficult environmental conditions.

3. Application of Concepts

For **real-time tracking in noisy environments**, robustness is extremely important.

Even if Method A initially provides better accuracy, its high sensitivity to noise can result in unstable motion estimates. This may lead to incorrect object trajectories or tracking errors.

Method B may produce slightly less precise estimates, but its low noise sensitivity means that it can maintain more consistent performance when the input contains noise.

For example, in surveillance footage with camera vibration or low-quality video, a noise-resistant method can be more useful than a highly accurate method that becomes unstable.

4. Conclusion

Method B is more suitable for real-time tracking in noisy environments.

Although Method A has higher accuracy, its high noise sensitivity makes it less reliable under noisy conditions. Method B offers moderate accuracy with low noise sensitivity, giving it better stability and robustness.

Therefore, for practical real-time tracking, Method B is preferred when noise is significant.

A possible improvement would be to combine the accuracy advantages of Method A with noise-reduction techniques such as filtering, smoothing, or robust optical-flow estimation.

14. Motion Estimation Accuracy vs Efficiency

Given Data

Method	Error	Computation
A	4%	High
B	9%	Low
C	6%	Moderate

1. Understanding of Data

The results show an inverse relationship between estimation error and computational requirements.

- **Method A:** Lowest error of 4%, but high computation.
- **Method B:** Highest error of 9%, but low computation.
- **Method C:** Error of 6% with moderate computation.

Thus, Method A prioritizes accuracy, Method B prioritizes efficiency, and Method C provides a compromise between both.

2. Analysis

For embedded systems, computational resources such as:

- CPU power
- Memory
- Energy
- Processing time

are often limited.

Method A provides excellent accuracy because its error is only 4%. However, high computational requirements can increase processing time and energy consumption. This may make it unsuitable for resource-constrained embedded devices.

Method B requires the least computation, making it highly efficient. However, its 9% error is considerably higher, which may reduce the reliability of motion estimation.

Method C has a 6% error and moderate computational requirements. It sacrifices only 2 percentage points of accuracy compared with Method A while requiring less computation.

3. Application of Concepts

For embedded systems, the best solution should consider both accuracy and computational efficiency.

Method	Accuracy	Efficiency	Suitability
A	Highest	Low	Accuracy-critical systems
B	Lowest	Highest	Highly resource-constrained systems
C	Good	Good	Balanced embedded systems

Method C provides the best compromise. Its 6% error is significantly better than Method B's 9%, while its moderate computation is more practical than Method A's high computation.

4. Conclusion

Method C is the best overall approach for embedded systems.

It provides a good balance between estimation accuracy and computational efficiency. Method A should be selected only when very high accuracy is essential and sufficient computational resources are available. Method B is appropriate when computational resources are extremely limited and higher error can be tolerated.

Therefore, Method C offers the most practical accuracy–efficiency trade-off for a typical embedded system.

15. GMM Performance in Different Conditions

Given Data

Scenario	Accuracy
Static	92%
Dynamic	68%

1. Understanding of Data

The Gaussian Mixture Model (GMM) performs very differently under static and dynamic conditions.

- In a **static environment**, GMM achieves **92% accuracy**.
- In a **dynamic environment**, accuracy decreases to **68%**.

The difference is:

$$92\% - 68\% = 24 \text{ percentage points}$$

This is a significant performance reduction.

2. Analysis

GMM-based background modelling generally works well when the background remains relatively stable. In a static environment, the model can learn the background distribution effectively, resulting in high detection accuracy.

However, dynamic environments contain continuous changes such as:

- Moving trees and leaves
- Changing shadows
- Moving background objects
- Changing illumination

- Camera movement
- Reflections
- Periodic motion

These changes can be incorrectly classified as foreground objects or can cause actual objects to be incorporated into the background model.

The reduction from **92% to 68%** indicates that the model is strongly affected by environmental dynamics.

3. Application of Concepts

The results show that GMM **cannot be considered equally reliable for all environments**.

Its 92% accuracy in static environments makes it highly useful for applications such as fixed-camera surveillance with relatively stable backgrounds.

However, its 68% accuracy in dynamic environments indicates that additional techniques are required.

Possible improvements include:

1. **Adaptive background modelling** – continuously update the background according to environmental changes.
2. **Shadow removal** – distinguish shadows from actual foreground objects.
3. **Noise filtering** – remove small false detections.
4. **Temporal filtering** – use information from multiple frames to improve stability.
5. **Camera-motion compensation** – compensate for movement of the camera.
6. **Hybrid approaches** – combine GMM with more advanced motion or object-detection techniques.

4. Conclusion

GMM is not equally reliable in all environments.

It performs very well in static conditions, achieving **92% accuracy**, but its performance drops significantly to **68% in dynamic environments**. The 24-percentage-point decrease demonstrates that environmental changes strongly affect GMM-based background modelling.

Therefore, GMM is **highly suitable for static environments but requires improvements for dynamic environments**. Adaptive background modelling, shadow removal, temporal filtering, and hybrid detection techniques can improve its robustness.

Overall, GMM should be selected according to the operating environment rather than being treated as a universally reliable method.